Python and Data Analytics Interview Questions

Q) Mention five benefits of using Python?

Ans: Python comprises of a huge standard library for most Internet platforms like Email, HTML, etc.

Python does not require explicit memory management as the interpreter itself allocates the memory to new variables and free them automatically

Provide easy readability due to use of square brackets Easy-to-learn for beginners

Having the built-in data types saves programming time and effort from declaring variables

Q) Name some type of conversion in Python

The term type of conversion used in Python means the conversion of one data type to another data type. For example:

int()– It is used to convert any data type into integer type

hex() – When there is a need to convert integers into hexadecimal, then this command is used.

Complex (real, image) –It is used to convert real numbers into complex number.

ListList () – It is needed when there is a need for conversion of any data type to a list type.

dict() – It is used for the conversion of a tuple of order into a dictionary.

float() – It is used when there is a need to convert any data type into float typeset() – It is used to convert nature into the set.

ord() – It easily converts characters into integer

oct() – It is used to convert an integer into octal

str() – It means the task of converting an integer into a string.

Q) In Python, do we use OOPs concepts?

The fact, Python is an object-oriented programming language where programs can solve by establishing an object model. But Python is treated as a structural and procedural language.

Q) Tell about Dict and List comprehensions

These are syntax constructions used to make the creation of a dictionary or list process easier that is based on existing iterable.

Q) Is python statically typed or dynamically typed?

Ans: Dynamic.

In a statically typed language, the type of variables must be known (and usually

declared) at the point at which it is used. Attempting to use it will be an error. In a dynamically typed language, objects still have a type, but it is determined at runtime.

You are free to bind names (variables) to different objects with a different type. So long as you only perform operations valid for the type the interpreter doesn't care what type they actually are.

Q) What are the rules for local and global variables in Python?

Ans: If a variable is defined outside function then it is implicitly global. If variable is assigned new value inside the function means it is local. If we want to make it global we need to explicitly define it as global. Variable referenced inside the function are implicit global.

Q) What Is the Difference Between an Iterator and Iterable?

Ans: The collection type like a list, tuple, dictionary, and set are all iterable objects whereas they are also iterable containers which return an iterator while traversing.

Q) What makes range and xrange in Python different?

Ans: xrange and range provide a way to generate a list of integers for your requirements. While range returns a Python list object, xrange returns an x range object which makes it ideal for generating a wide range of list.

Q) What is a dictionary in python?

Ans: Dictionary is a data structure as well as a data type in python. It is enclosed in curly brackets{}.

Dictionary contains 2 elements – key and value

key is a string for us to grab a value.

Q) What Is the Index in Python?

Ans: An index is an integer data type which denotes a position within an ordered list or a string.

In Python, strings are also lists of characters. We can access them using the index which begins from zero and goes to the length minus one.

For example, in the string “Program,” the indexing happens like this:

Program 0 1 2 3 4 5

Q) What are Python decorators?

Ans: A specific change made in Python syntax to alter the functions easily are termed as Python decorators.

Q) What are generators in Python?

Ans: The way of implementing iterators are known as generators. It is a normal function except that it yields expression in the function.

Q) How do you do data abstraction in Python?

Ans: An abstraction means hiding away information or showing only information that’s necessary.

Q) Differentiate between list and tuple.

Ans: Tuple is not mutable it can be hashed eg. key for dictionaries. On the other hand, lists are mutable.

Q) Define slicing in Python.

Ans: Slicing refers to the mechanism to select the range of items from sequence types like lists, tuples, strings.

Q) Define self in Python.

Ans: An instance of a class or an object is self in Python. It is included as the first parameter. It helps to differentiate between the methods and attributes of a class with local variables.

Q) What is pep 8?

Ans: Python Enhancement Proposal or pep 8 is a set of rules that specify how to format Python code for maximum readability.

Q) What is GIL?

Ans: GIL or the Global Interpreter Lock is a mutex, used to limit access to Python objects. It synchronizes threads and prevents them from running at the same time.

Q) What is encapsulation?

Ans: Encapsulation is the binding of data and functions that manipulate the data.

It is a process of wrapping up data and variables together.

Q) What is Polymorphism in Python?

Ans: Polymorphism means the ability to take multiple forms. So, for instance, if the parent class has a method named ABC then the child class also can have a method with the same name ABC having its own parameters and variables. Python allows polymorphism.

Q) Does python make use of access specifiers?

Ans: Python does not deprive access to an instance variable or function. Python lays down the concept of prefixing the name of the variable, function or

method with a single or double underscore to imitate the behaviour of protected and private access specifiers.

Q) How are arguments passed by value or by reference?

Ans: Everything in Python is an object and all variables hold references to the objects. The references values are according to the functions; as a result, you cannot change the value of the references. However, you can change the objects if it is mutable.

Q) What are Python libraries? Name a few of them.

Ans: Python libraries are a collection of Python packages. Some of the majorly used python libraries are – NumPy, Pandas, Matplotlib, Scikit-learn and many more.

Q) What is module and package in Python?

Ans: In Python, module is the way to structure program. Each Python program file is a module, which imports other modules like objects and attributes.

The folder of Python program is a package of modules. A package can have modules or subfolders.

Q): What is the python "with" statement designed for?

Ans: The with statement simplifies exception handling by encapsulating common preparation and clean-up tasks in so-called context managers.

For instance, the open statement is a context manager in itself, which lets you open a file, keep it open as long as the execution is in the context of the with statement where you used it, and close it as soon as you leave the context, no matter whether you have left it because of an exception or during regular control flow.

Q) How will you differentiate between deep copy and shallow copy?

Answer: We use a shallow copy when a new instance type gets created. It keeps the values that are copied in the new instance. Just like it copies the values, the shallow copy also copies the reference pointers.

Reference points copied in the shallow copy reference to the original objects. Any changes made in any member of the class affect the original copy of the same. Shallow copy enables faster execution of the program.

Deep copy is used for storing values that are already copied. Unlike shallow copy, it doesn't copy the reference pointers to the objects. Deep copy makes the reference to an object in addition to storing the new object that is pointed by some other object.

Changes made to the original copy will not affect any other copy that makes use of the referenced or stored object. Contrary to the shallow copy, deep copy makes execution of a program slower. This is due to the fact that it makes some copies for each object that is called.

Q) What do you understand by monkey patching in Python?

Answer: The dynamic modifications made to a class or module at runtime are termed as monkey patching in Python.

Q) What is inheritance?

When a class uses methods constructed under another class then it is defined a s inheritance.

Classes termed as child or subclass inherit methods and variables from parent class or base class.

Q) Which statement is used when no action is required in the program except requiring syntactically?

The statement which should be used is "pass." It is important that you use the same in the lowercase since statements in Python are case sensitive.

Q) Difference between pickling and unpickling?

Any Python Interview Question and Answers guide won't complete without this question. In Python, the pickle module accepts any Python object, transforms it

into a string representation, and dumps it into a file by using the dump function. This process is known as pickling. The function used for this process is pickle.dump().

On the other hand, the process of retrieving the original Python object from the stored string representation is called unpickling. The function used for this process is pickle.load().

Q) What is the map() function used for in Python?

Ans: The map() function applies a given function to each item of an iterable. It then returns a list of the results. The value returned from the map() function can then be passed on to functions to the likes of the list() and set().

Typically, the given function is the first argument and the iterable is available as the second argument to a map() function. Several tables are given if the function takes in more than one arguments.

Q) Whenever Python exits, all the memory isn't deallocated. Why is it so?

Ans: Upon exiting, Python's built-in effective cleanup mechanism comes into play and try to deallocate or destroy every other object.

However, Python modules that are having circular references to other objects or the objects that are referenced from the global namespaces aren't always deallocated or destroyed.

This is because it is not possible to deallocate those portions of the memory that are reserved by the C library.

Q) Explain memory managed in Python?

Ans: Python private heap space takes place of memory management in Python. It contains all Python objects and data structures. The interpreter is responsible to take care of this private heap and the programmer does not have access to it. The Python memory manager is responsible for the allocation of Python heap space for Python objects. The programmer may access some tools for the code with the help of the core API. Python also provides an inbuilt garbage collector, which recycles all the unused memory and frees the memory and makes it available to heap space.

Q) What is the lambda function?

Ans: An anonymous function is known as a lambda function. This function can have only one statement but can have any number of parameters.

Q) What is the working of a break, pass, and continue?

Breaks: It permits loop termination at the time some condition is fulfilled, and the control is sent to the next statement.

Pass: It is used when the users require some block of code syntactically; however, the users seek to skip the execution. Generally, it is a null operation. During its performance, nothing happens.

Continue: It permits skipping some phase of a loop after fulfilment of a particular condition and also when control is sent to the beginning of the loop.

Q) What do you know about the unit test in Python?

United means a unit testing framework in Python. It supports the sharing of automation testing, aggregation of tests, setups, etc.

Q) What are Python namespaces? Why are they used?

Ans: A namespace in Python ensures that object names in a program are unique and can be used without any conflict. Python implements these namespaces as dictionaries with 'name as key' mapped to a corresponding 'object as value'. This allows for multiple namespaces to use the same name and map it to a separate object. A few examples of namespaces are as follows:

Local Namespace includes local names inside a function. the namespace is temporarily created for a function call and gets cleared when the function returns.

Global Namespace includes names from various imported packages/ modules that is being used in the current project. This namespace is created when the package is imported in the script and lasts until the execution of the script.

Built-in Namespace includes built-in functions of core Python and built-in names for various types of exceptions.

Lifecycle of a namespace depends upon the scope of objects they are mapped to. If the scope of an object ends, the lifecycle of that namespace comes to an end. Hence, it isn't possible to access inner namespace objects from an outer namespace.

Q) What are the tools that help to find bugs or perform static analysis?

Ans: PyChecker is a static analysis tool that detects the bugs in Python source code and warns about the style and complexity of the bug. Pylint is another tool that verifies whether the module meets the coding standard.

Q) How to delete a file in Python?

Ans: By using an authority os.remove (filename) or os.unlink (filename)

Q) What is Scope in Python?

Ans: Every object in Python functions within a scope. A scope is a block of code where an object in Python remains relevant. Namespaces uniquely identify all the objects inside a program. However, these namespaces also have a scope defined for them where you could use their objects without any prefix. A few examples of scope created during code execution in Python are as follows:

• A local scope refers to the local objects available in the current function.

• A global scope refers to the objects available throught the code execution since their inception.

• A module-level scope refers to the global objects of the current module accessible in the program.

An outermost scope refers to all the built-in names callable in the program. The objects in this scope are searched last to find the name referenced.

Q) What is __init__() in Python?

Ans: Equivalent to constructors in OOP terminology, __init__ is a reserved method in Python classes. The __init__ method is called automatically whenever a new object is initiated. This method allocates memory to the new object as soon as it is created. This method can also be used to initialize variables.

Q) Why is <__init__.py> module used in Python?

The <__init__.py> module can help in fulfilling the following objectives.

• It makes Python interpret directories as containing packages by excluding the ones with a common name such as string.

• It grants a programmer with the control to decide which directory is a package and which is not.

• However, the <__init__.py> can also be an empty file. It can then help in executing the initialization code for a package or setting the <__all__> variable.

Q) What does this stuff mean: *args, **kwargs? And why would we use it?

Ans: We can use *args when we are not sure how many args we are going to pass to a function. We can also use if we want to pass a stored list or tuple of arguments to a function. **kwargs is used when we don't know how many keyword arguments will

be passed to a function, or it can be used to pass the values of a dictionary as keyword arguments. The *args and **kwargs are a convention. You can also use *abc and **wxyz .

Q) What Does The "Self" Keyword Do?

Ans: The self is a Python keyword which represents a variable that holds the instance of an object.

In almost, all the object-oriented languages, it is passed to the methods as a hidden parameter.

Q) What is zip function?

Ans: Python zip() function returns a zip object, which maps a similar index of multiple containers. It takes an iterable, converts into iterator and aggregates the elements based on iterables passed. It returns an iterator of tuples.

Q) What is PIP?

PIP is an acronym for Python Installer Package which provides a seamless interface to install various Python modules. It is a command-line tool that can

search for packages over the internet and install them without any user interaction.

Q) How to debug a Python program?

Ans: Use the following command to debug Python program:

$ python -m pdb python-script.py

Q) How is Multithreading achieved in Python?

Ans:

1. Python has a multi-threading package but if you want to multi-thread to speed your code up, then it's usually not a good idea to use it.

2. Python has a construct called the Global Interpreter Lock (GIL). The GIL makes sure that only one of your 'threads' can execute at any one time. A thread acquires the GIL, does a little work, then passes the GIL onto the next thread.

3. This happens very quickly so to the human eye it may seem like your threads are executing in parallel, but they are really just taking turns using the same CPU core.

4. All this GIL passing adds overhead to execution. This means that if you want to make your code run faster then using the threading package often isn't a good idea.

Q) What Are Errors And Exceptions In Python Programs?

Ans: Errors are coding issues in a program which may cause it to exit abnormally.

On the contrary, exceptions happen due to the occurrence of an external event which interrupts the normal flow of the program.

Q) How Do You Handle Exceptions With Try/Except/Finally In Python?

Ans: Python lay down Try, Except, Finally constructs to handle errors as well as Exceptions. We enclose the unsafe code indented under the try block. And we can keep our fall-back code inside the except block. Any instructions intended for execution last should come under the finally block.